

Thibault Verrue

Junior AI Engineer



WORK EXPERIENCE

Cluma Rolbruggen, Roeselare – AI Engineer

July 2026 - August 2026

- Continued development of my AI internship project toward a production-ready deployment.
- Improved system reliability, performance, and maintainability for real-world use.
- Collaborated with engineers to integrate the AI solution into existing company workflows.

Cluma Rolbruggen, Roeselare – AI Intern

Februari 2026 - June 2026

- Implementing a document-grounded LLM system using RAG architecture with embeddings and semantic search
- Processing structured and unstructured technical data (PDF, Excel, ERP) into a unified vector knowledge base
- Developing a Python-based backend API and interactive chat UI for real-time technician support
- Exploring deployment strategies for on-premise and cloud AI inference
- Delivering a production-ready prototype improving technician information retrieval speed

Barco, Kortrijk — Self Supervised AI Training

July 2025 - September 2025

- Trained self-supervised vision models on medical pathology image datasets using PyTorch
- Evaluated downstream classification performance and optimized data pipelines in a Linux GPU environment

Bekaert, Zwevegem — IT Intern

May 2023- June 2023

- Installing new desktops / docking stations for employees.
- Converting Python scripts to .exe files..

DETAILS

Demeesterstraat 6
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Driver's License B

IT-skills

AI:

- Machine Learning
- Deep Learning
- Reinforcement Learning
- NLP

Computer Vision & Edge AI

- OpenCV
- PyTorch/TensorFlow
- Image classification / detection
- Real-time inference
- Model optimisation (quantization, pruning, ONNX)

Embedded & Systems

- Raspberry Pi / LIMO Pro
- Linux / Bash
- Camera & sensor integration

Programming Languages

- Python
- Java
- C#

Other

- Docker
- GIT
- Azure
- Unity

LANGUAGES

EDUCATION

Creative Technologies and Artificial Intelligence, Howest Bachelor

September 2023 - August 2026

Native: Dutch

Advanced: English

Basic: French

IT Management, Guldensporencollege Kaai — TSO diploma

September 2020 - June 2023

Projects

Edge-Optimised Vision Model on LIMO Pro

November- December 2025

- Trained computer vision model for real-time inference
- Hand gesture recognition
- Applied pruning & quantization for embedded efficiency
- Deployed on LIMO Pro with Linux-based pipeline
- Validated latency and on-device performance

Smart Waste Sorting System (Computer Vision & IoT)

May-June 2024

- Computer vision project classifying waste types (PMD, paper, residual) using a custom dataset and deep learning model, integrated with Raspberry Pi hardware for automated bin control and real-time visualization.
- **Tech used:** Python · Computer Vision · Deep Learning · Raspberry Pi · Sensors · Motors · Camera systems · Real-time inference

Multi-Agent AI Python Tutor (Generative AI & Agentic Systems)

January 2026

- Curriculum-aware AI tutoring system deployed as a Discord bot, using multi-agent orchestration to deliver structured Python lessons, generate exercises, evaluate code, and adapt instruction based on learner progress.
- **Tech used:**
Python · Generative AI · LLMs · Multi-Agent Systems · LangGraph · LangChain · RAG · Vector Databases · Discord API · Prompt Engineering